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CONFIDENTIAL

MASTER FORMULA AND METHOD

OF

RINGER'S LACTATE INTRAVENOUS INFUSION BP 500ml

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BIOMEDICAL LIMITED

ILORIN

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Approval Page

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A. Product Information

Brand Name : Bioflex Intravenous Injection Ringer's Lactate

Generic Name : Ringer's Lactate Intravenous Infusion BP

Product Category : Isotonic Buffered Extracellular Fluid Replacement

Dosage Form : Injectable

Route of Administration : Intravenous

Label Claim : Each 100 ml Contains…

Sodium Lactate	BP	0.53gm
Sodium Chloride	BP	0.6gm
Potassium Chloride	BP	0.04gm
Calcium Chloride	BP	0.0275gm
Water for Injection	BP q.s.	to 100 ml

Pack Size : 500ml Pouch.

Packing Pattern : 20 pieces of over wrapped pouches packed in 5-ply cartons along with the handling and storage label. The cartons are sealed with gum or starch paste.

Shelf Life : 36 Months.

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C. Manufacturing Formula

Standard Batch Size = 1600 Liters.
Standard Fill Volume = 505ml per PVC Pouches.

Sr. No.	Raw Materials	Rational	Label claim/100ml	% Ove-rages	Std. Batch Qty.
1	Sodium Lactate	Active	0.53gm	Nil	8.5Kg
2	Sodium Chloride	Active	0.6gm	Nil	9.6Kg
3	Potassium Chloride	Active	0.04gm	Nil	0.64Kg
4	Calcium Chloride	Active	0.0275gm	Nil	0.44Kg
5	Water for Injection BP	Vehicle	N.A.		q.s. to 1600 Liters.

PRIMARY PACKING MATERIAL

Sr. No.	Name of the Material	Unit	Batch Qty.	% Excess	Std. Batch Qty
1	500 ml PVC Pouches	Nos.	3025	Nil	3025
2.	Stoppers	Nos.	3055	2%	3025

B. Manufacturing Conditions and Precautions

1. Strictly follow the procedures of Entry/Exit to Clean Areas.
2. Ensure that the manufacturing area and equipment are clean.
3. Ensure that at all the time current good manufacturing practices are followed.
4. Follow safety measures all the time.

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5. Ensure that the weighing scale is calibrated.
6. Verify the weight of dispensed raw material.
7. Rinse the equipment with fresh water for injection before every use.
8. Check the manufacturing area & Filling area for pre-operation checks like valve opening and closing.
9. Destroy the rejected PVC pouches as per the procedure.
10. Avoid product exposure to dirt during manufacturing, filling and storage.

SR. No	NAME OF EQUIPMENT	IDENTIFICATION No.	QUANTITY	CAPACITY
1	WEIGHING BALANCE	BL/WB/PE/001	1	50kg
2	DISPENSING BOOTH	BL/DSP/PE/001	1	FITTED WITH 0.2µ HEPA FILTER
3	MIXING TANK	BL /SPR/PE/OO7-3	1	2000 LITERS
4	MIXING TANK	BL /SPR/PE/OO7-2	1	2000 LITERS
5	MIXING TANK	BL /SPR/PE/OO7-1	1	2000 LITERS
6	SEMI AUTOMATIC FILLING MACHINE	BL/FR/PE/001 – 3	3	0 -2000ML
7	SEMI AUTOMATIC FILLING MACHINE	BL/FR/PE/004 – 5	2	0 -2000ML
8	DE-LAMA AUTOCLAVE (STERILISER)	BL/AC/PM/001	1	3,200Litres
9	PHARMALAB AUTOCLAVE	BL/AC/PM/002	1	4,200Litres
10	CLEAN ROOM	BL/CL/PE/001	1	FITTED WITH 0.2µ HEPA FILTER
11	POUCH CODING MACHINE	BL/BC/PE/001-004	4	NA

E. Manufacturing Procedures

1 Samples of Water for Injection B.P. are collected and send for analysis to the Quality Assurance department. If the Quality assurance certifies the water complies for manufacturing, proceed to step 2.

2 Turn on the ball valve the open WFI into the tank, switch on WFI transfer pump and allow WFI to run from the storage vessel into the calibrated manufacturing vessel up to 1400 liters of the required volume.

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3 Weigh 9.6Kg of Sodium Chloride, 8.5Kg of Sodium Lactate, 0.64Kg of Potassium Chloride, 0.44Kg of Calcium Chloride on the pre calibrated Weighing scale under the dispensing booth in the presence of Production Pharmacist and QA officer.
4 Transfer the dispensed Raw materials into the manufacturing Tank.
5 Note the time and record it.
6 Turn on the stirrer and mix thoroughly, gradually making up the volume to 1600Liters.
7 Continue stirring for 10 – 15minutes to dissolve the ingredient.
8 Inform the Quality Assurance so that samples can be collected for analysis.
9 If Quality Assurance has certified the bulk solution to be within the compendia specification, intimate the filling section so that filling can commence.
10 Ensure solution transfer pump is switched "ON"
11 Fix the sterile filter membranes in the filter holding unit
12 Fix the 0.8µ pre filter and 0.45µ membrane filters in the 293mm filter holder and 0.2µ cartridge filter into the 10'' filter holder and ensure that there are no leakages of solution during filtration.
13 Check the correctness of the printing details of the pouches. Record the filtration and filling details in the Batch Record.
14 Adjust the fill volume 505ml for 500ml pouch filling.
15 Rinse the stoppers in WFI and keep near the filling machine
16 Switch on the filling machine and fill the required volume in PVC pouches and insert Stopper inside the open neck of the PVC pouch and pass through the transfer box to loading area for sterilization
17 Check the fill volume by measuring with a measuring cylinder the volume of the filled pouch every 10 minutes from the filling nozzles and record it in Batch Record.
18 Follow proper cleaning procedure batch to batch and product to product cleaning.
19 Sterilize the filled pouches in super-heated water spray sterilizer at 116°C-118°C for 40 min
20 QA will take 4 samples for finished product analysis from the sterilized product on the trolleys.
21 Transfer the batch for visual inspection and packing.

F. In process Control

Sr. No.	Stage	Checks
1.	Dispensing of raw material.	1. Line Clearance 2. Balance calibration. 3. Release status of raw material from Q.C. 4. QC Reference Number, Quantity. 5. Cleanliness of dispensing equipment's, dispensing containers/Nylon wrapper.

2.	Manufacturing of bulk & filtration	1. Line Clearance 2. Area cleanliness. 3. Equipment & machinery cleanliness 4. Addition of active ingredients. 5. Pre-operation Checks 6. Volume of solution. 7. pH 8. Clarity of solution 9. Filtration process. 10. Bulk analysis and release status from Q.C. 11.Environment control
3.	Filling & Sealing.	1. Q. C. release status for filling. 2. Pre-operation checks. 3. Area cleanliness. 4. Volume of filled pouches. 5. Average fill volume. 6. Fill volume variation. 8. Sealing quality.
4.	Quarantine	1. Pre-operation checks. 2. Storage of off loaded pouches on packing trolleys with proper status label.

G. Packaging Formula

Standard Batch Size = 1600 Liters for 500ml pack size.

Sr. No.	Name of material	Unit	Quantity Required	% Ove-rages	Std. Batch Qty.
1	Nylon wrappers 500ml	No	3055	1 %	3025*
2	500ml Infusion Carton	No	150	Nil	150
3	Stickers	No	150	Nil	150

***** Subjected to variation depending on yield at visual inspection stage.

Packing Pattern: 20 pieces of over wrapped pouches packed in 5-ply cartons along with the handling and storage label. The cartons are sealed with gum or starch paste.

H. Packaging Conditions and Precautions.

1. Check the cleanliness of area & equipment.
2. Check for line clearance & record the details in Batch Record.
3. Ensure that all packing materials required for a batch are requested.
4. Verify all printed matters on the pouch such as:

i) Batch No.
ii) Mfg. Date.
iii) Exp. Date.

5. Ensure withdrawal of retention sample.

I. Packaging Procedure and In-Process control.

(Visual Inspection and packing is a simultaneous continuous operation)

1. Perform the pre-operation checks. Ensure for Q.A. release is obtained.
2. Check for the line for proper arrangement of workers.
3. On the carton check the proper marking roller fix the required digits and alphabets stating product name, batch number, Mfg. Date & Exp. date.
4. Take one trolley of the batch to the packing line.
5. Intimate the In-Process Quality Assurance Inspector for the line clearance.
6. In Process Quality Assurance Inspector will check for the line clearance & sign in the specified place in the BMR.
7. Start coding of cartons & packing of batch.
8. Receive the batch from Visual inspection area (On line visual inspection and packing)
9. Put the pouches to LDPE bags with the tube side of the pouch to the inner side of the bag without tearing the bag.
10. Transfer the wrapped pouches to the cartooning area.
11. Put the wrapped pouches in the cartons on which a product identification sticker label has been affixed.
12. Arrange 20 pouches in a 5-ply carton.
13. The arrangement of pouches in the batch coded carton having product identification sticker affixed should be of uniform pattern & should be placed flatly in the cartons.
14. Seal the cartons with gum or starch paste freshly prepared. Apply the gum or starch paste with a brush.
15. 10 pouches for QA retention samples per batch taken for reference.

J. Disposal of Rejects.

1. Any rejected pouches from the batch coding, Filling, Sterilization and Packing sections will be collected by the supervisor every shift & will get approval from Quality Assurance Manager and Superintendent pharmacist for destruction on batch wise.
2. The Supervisor when handing over the rejected pouches for destruction will write the details of the rejections in the record & get it countersigned by the reject destruction personnel as well as QA personnel.

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3. The rejected pouches are slitted at least to a span of 3 inches with the help of blade scissors & drained out if there is any solution remaining in the pouch. This operation will be carried out on direct supervision of the QA personnel.
4. After this slitted pouches are packed into LDPE bags and transferred to scrap yard for sale to the Authorized scrap collection contractor.

K. Yield Analysis

Standard Batch size = 1600 Liters.

Sr. No.	Stage	Standard Yield %	Permissible Yield %
1.	Batch Coding	100%	NLT 99.50%
2.	Filling	100%	NLT 99.50 %
3.	Packing	100%	NLT 94%

L Reprocessing

Batch reprocessing is not practiced.

M Process Deviation

Any inevitable deviation should not be carried out without approval of Quality Assurance Manager and authorization of Superintendent Pharmacist. Record any process deviation in a specified format in the deviation form and attach same to the BMR.

N Raw Material Specification

1. Ringer's Lactate (As per British Pharmacopoeia)

O Packing Material Specification

1. PVC pouches

2. Stoppers
3. 500 ml Nylon Wrappers
4. 500 ml Cartons
5. Gum or Starch paste
6. Infusion leaflets/Labels

Q Finish Product Specification

QC Specification (As per British Pharmacopoeia)

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